

Assignment 1: Sorting visualiser

Build a small visualiser that shows insertion sort and mergesort rearranging an array of bars.

This is an individual-or-pair assignment. Pair work is allowed. Submit one repository and name both authors in the README.

What to submit

- Source code that runs without undocumented steps.
- A 60-second screen recording of both algorithms on the same shuffled input.
- A short write-up (one page) comparing comparison counts on $n = 20$, 50, and 100.

Submit before 23:59 on 14 October 2026. The portal timestamp is the one we use. Email is not a submission.

Late work loses 10% per day. After three days we stop accepting it.

Requirements

1. The user can generate a random array of n bars, n between 8 and 80.
2. Play, pause, and step-once controls.
3. Insertion sort and mergesort, selectable, sharing the same drawing code.
4. A counter of comparisons, updated live.

You do not need a framework. Canvas or SVG is enough. If you use a library, say why in the README.

Grading

Correctness of the two sorts is 50%. The animation being readable (not a blur of swaps) is 30%. The write-up is 20%. A visualiser that only works for $n = 8$ will not pass.

Questions go in the course forum, not in the commit history.